Entanglement Distribution Protocol in a Linear Quantum Repeater Network

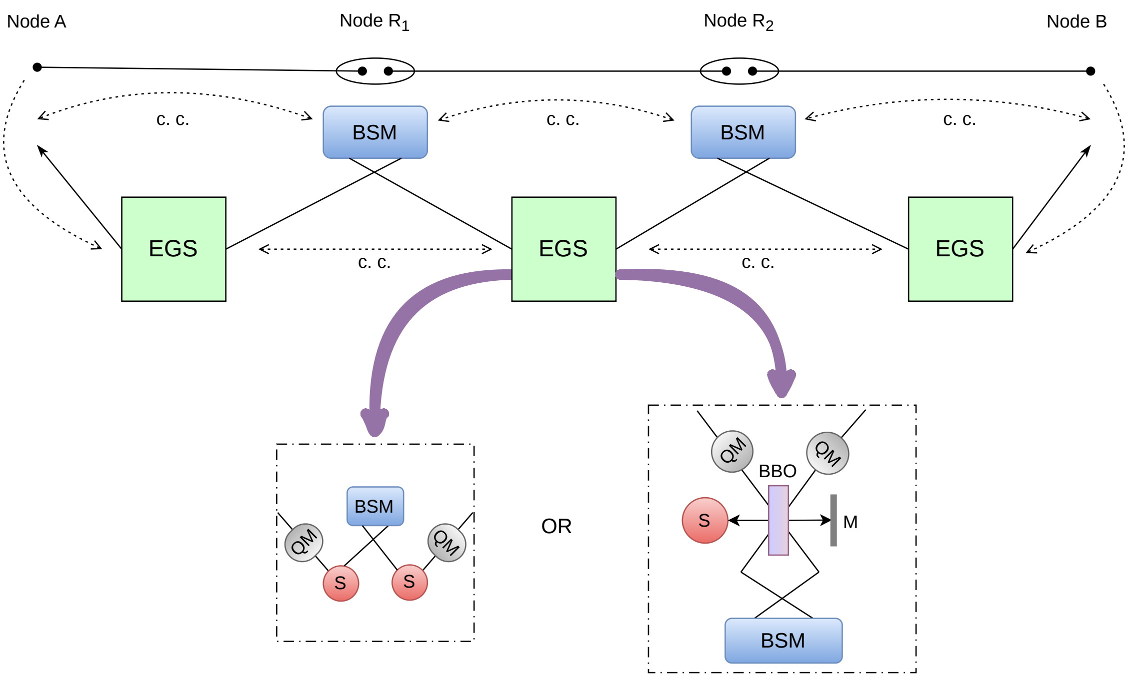

Figure 1: Illustration of the proposed EGS-assisted quantum repeater architecture. Neighboring nodes are connected through dedicated Entanglement Generation Sources (EGSs), which perform heralded elementary-link generation using one of the two possible implementations shown in the figure. Upon successful link establishment, heralding information is transmitted via classical communication channels to the repeater nodes, where Bell-state measurements are performed to realize entanglement swapping and establish end-to-end entanglement between the communicating parties.

1 Quantum Repeater Protocol

Quantum repeaters are essential components of large-scale quantum networks, enabling long-distance entanglement-based quantum communication. Due to photon loss and decoherence in optical channels, direct distribution of entanglement over long distances becomes highly inefficient. Quantum repeaters overcome this limitation by dividing the communication channel into shorter elementary links and establishing entanglement between neighboring repeater nodes. Through entanglement swapping operations performed at intermediate nodes, entanglement can subsequently be extended between distant communicating parties.

In this section, we briefly describe a protocol for entanglement distribution over a linear network.

Stage 1: Elementary Link Generation

An Entanglement Generation Source (EGS) is placed between every pair of adjacent nodes in the network. Each EGS consists of either two SPDC sources, two quantum memories (QMs) and a Bell-state measurement (BSM) module or one SPDC source, two QMs and a BSM module.

Both architectures realize *heralded elementary-link generation*. In the first implementation, the BSM performed on photons generated by the two independent SPDC sources heralds the successful establishment of entanglement between the quantum memories. In the second implementation, the BSM similarly confirms successful loading and establishment of the elementary entangled link.

Whenever the BSM succeeds, the EGS declares successful elementary-link establishment and communicates this heralding information to the neighboring EGS nodes through classical communication channels. The corresponding quantum states are retained in the quantum memories and become available for the subsequent entanglement swapping stage.

Stage 2: Entanglement Swapping at Repeater Nodes

Each repeater node is equipped solely with a BSM module only. After receiving heralding signals from the adjacent EGSs, the corresponding photons stored in the quantum memories are released and transmitted toward the repeater node. The repeater node then performs the BSM on the received photons. A successful BSM extends entanglement across multiple elementary links through entanglement swapping. The outcome of the measurement is subsequently communicated to the neighboring nodes via classical channels.

Stage 3: Entanglement Verification

The classical information generated by all EGSs and repeater nodes is distributed throughout the network. Using this shared information, the network nodes collectively determine whether a continuous chain of successful elementary-link generations and entanglement-swapping operations exists between the communicating end nodes. When such a chain is identified, an end-to-end entangled state is considered to have been successfully established between the source and destination nodes. This entangled resource can subsequently be utilized for quantum communication applications such as quantum key distribution, quantum teleportation, and distributed quantum computing.

The proposed architecture separates the tasks of entanglement generation and entanglement swapping by assigning them to dedicated EGSs and repeater nodes, respectively. This modular design simplifies network operation provides explicit heralding information for link establishment.

2 Protocol Summary

Algorithm 1: EGS-Based Entanglement Distribution Protocol

```
foreach Entanglement Generation Source (EGS) do
    Attempt heralded elementary-link generation between adjacent nodes;
    if elementary link is successfully established then
        Store the entangled qubits in the corresponding quantum memories (QMs);
        Notify the adjacent repeater nodes through classical communication;
    else
        Repeat the entanglement generation process;
foreach Repeater Node do
    Wait until successful link-establishment messages are received from both adjacent
      EGSs;
    Release the corresponding qubits from the quantum memories;
    Perform Bell-state measurement (BSM) to realize entanglement swapping;
    Broadcast the measurement outcome through classical communication channels;
Collect the heralding information from all EGSs and the BSM outcomes from all
  repeater nodes;
Determine whether a continuous chain of successful elementary links and
  entanglement-swapping operations exists between the communicating end nodes;
if an end-to-end entangled path is identified then
    Declare successful end-to-end entanglement establishment;
    Reserve the entangled state for quantum communication tasks;
else
    Repeat the protocol in the next communication round;
```

References

[1] X. Liu, J. Hu, Z.-F. Li, X. Li, P.-Y. Li, P.-J. Liang, Z.-Q. Zhou, C.-F. Li, and G.-C. Guo, "Heralded entanglement distribution between two absorptive quantum memories," *Nature*, vol. 594, no. 7861, pp. 41–45, 2021.

[2] C. Mesny, F. Guillemin, and C. Goursaud, "Entanglement distribution protocols under imperfect fidelity and quantum memory conditions," *arXiv preprint arXiv:2605.31347*, 2026.